Chapter 1

Theorem 1. let $\rho : G \to GL(V)$ be a representation of G in V, and let $W \subset V$ be a vector subspace of V stable under G. Then there exists a complement W^0 of W in V which is stable under G.

Proof. Let W^0 be a complement of W in V, and let $p : V \to V$ be the projection of V onto W corresponding to the decomposition $V = W \oplus W^0$. Let $s \in G$. We define a new projection $p^0 : V \to V$:

$$p^0 v := \frac{1}{|G|} \sum_{s \in G} \rho_s p \rho_s^{-1} v.$$

Notice that p maps to W and W is G-invariant, so $\rho_s p$ sends elements to W. If $w \in W$, then $\rho_s^{-1} w \in W$. Hence,

$$p^0 w = \frac{1}{|G|} \sum_{s \in G} \rho_s p \rho_s^{-1} w = \frac{1}{|G|} \sum_{s \in G} \rho_s \rho_s^{-1} w = w.$$

So, p^0 is a projection of V to W.

Let $x \in W^0$. Then, $p^0 \rho_s x = \frac{1}{|G|} \sum_{s \in G} \rho_s p \rho_s^{-1} \rho_s x = \frac{1}{|G|} \sum_{s \in G} \rho_s p x = 0$. Since $p^0(\rho_s x) = 0$ for any $x \in W^0$, we have $\rho_s x \in W^0$. Hence, W^0 is stable under G. $\square$